MILITARY SPECIFICATION

CARTRIDGE, 5.56MM, REFERENCE

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Cartridge, 5.56mm, Reference to be used for calibrating acceptance testing equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspections of Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1305

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MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-636	-	Visual Inspection Standards for Small Arms Ammunition Through Caliber .50
MIL-STD-644	-	Visual Inspection Standards andInspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-1168	-	Lot Numbering of Ammunition

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

7643674	-	Classification of Cartridge Case Defects for Small Arms Ammunition
10524633	-	Cartridge, 5.56mm, Reference
10523634	-	Packing and Marking, Cartridges, 5.56mm, Cartons, Box, Ammunition, M2A1
10523635	-	Packaging and Marking, Cartridges, 5.56mm, Cartons, Box, Ammunition, M2A1, Box, Wirebound
LI-10524633	-	Index of Inspection Equipment List for Cartridges 5.56mm, Reference

PUBLICATIONS

SCATP-5.56mm	-	Ammunition Ballistic Acceptance Test Methods Test Procedures for 5.56mm Cartridges

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Other publications. The following documents form a part of this specification to the extent specified herein in the issue of documents which are indicated as DOD adopted shall be the issue listed in the current DoDISS and supplement thereto if applicable:

ASTM E92	-	Standard Test Method for Vickers Hardness of Metallic Materials

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CODE OF FEDERAL REGULATIONS

49 CFR-100-199 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosive and Other Dangerous Article

(The Interstate Commerce Commsision regulations are now part of the Code of Federal Regulations (1967 editions and revisions) available from the Superintendent of Documents, Government Printing Office, Washington, DC 20315. Orders for the above publications, cite 49CFR-100-199 (latest revision).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The cartridge shall comply with drawing 10524633, referenced specifications and the following:

3.2 Bullet extraction. The force necessary to extract the bullet from the cartridge case shall be not less than 35 pounds.

3.3 Residual stress. The cartridge case shall not split when subjected to one percent mercurous nitrate solution for 15 minutes.

3.4 Waterproofness. The cartridge shall not release more than one bubble of air when subjected to a positive internal pressure of 7.5 pounds per square inch (psi) for 30 seconds.

3.5 Velocity. The grand average of the velocities of all the sample cartridges, conditioned at 70° Fahrenheit (F), shall be 3250 feet per second (ft/sec) plus or minus 20 ft/sec, at 78 feet from the muzzle of the weapon. The average of the standard deviations of the velocities for 15 20-round series shall not exceed 25 ft/sec.

3.6 Chamber pressure. Measurement to be made by piezoelectric transducer. The average chamber pressure of the sample cartridges, conditioned at 70°F shall not exceed 55,000 psi. The average of the standard deviations of chamber pressures for 15 20-round series shall not exceed 2,000 psi.

3.7 Port pressure. Measurement to be made by piezoelectric transducer. The average mean port pressure of the sample cartridges conditioned at 70°F shall be 14,400 PSI ± 1,000 psi.

3.8 Function and casualty. The cartridge shall function without casualty when conditioned at 70°F.

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3.9 Workmanship. The metal parts of the cartridge shall be free of cracks, splits, perforations, burrs and foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by any cleaning agent. All parts and assemblies shall be fabricated, loaded and assembled in a thorough and workmanlike manner. In addition, the cartridge shall comply with the standards specified in the 7.62mm section of MIL-STD-636. Extreme care should be exercised to avoid contamination of primers or propellant by oil, grease or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

Quality Conformance Inspection (see 4.3)

4.3 Quality Conformance Inspection

4.3.1 Lot formation. In accordance with MIL-A-48078.

4.3.1.1 Lot submission. The product shall be submitted in accordance with MIL-STD-105. The entire lot of reference cartridges shall be assembled and charged on a single loading machine. (See 6.7)

4.3.1.2 Component parts. Unless otherwise specified, component lots shall be homogeneous and of a size convenient to the contractor and inspected, tested and accepted by the contractor. The cartridge lot shall not contain:

a. Cartridge cases from more than one manufacturer.

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b. Primers from more than one lot interfix number from one manufacturer.

c. Bullets from more than one interfix number from one manufacturer.

d. Propellant from more than one lot and more than one manufacturer.

4.3.1.3 Lot identification. Each lot of ammunition shall be identified as to type, caliber and model, as well as a lot number in accordance with MIL-STD-1168.

4.3.2 Examination for defects.

a. Major and minor defects - Examintion for major and minor defects shall be performed on a class basis in accordance with classification of defects using applicable sampling plans and acceptance criteria of MIL-STD-105. The acceptable quality level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent unless otherwise specified. All non-conforming cartridges shall be rejected.

b. Critical defects - Unless otherwise specified, one hundred percent examination shall be performed for all critical defects. If a visual critical defect is found in a sample either just prior to a firing test or after a firing test (and the defect is not due to the firing), the lot shall be rejected. The lot shall then be rescreened and resubmitted for visual inspecton of critical defects. If a critical defect is found during packing, the portion of the lot that has been packed or is in the process of being packed shall be rejected. In addition, the portion of the lot remaining to be packed shall be rejected. The lot shall then be rescreened and resubmitted for visual inspection of critical defects.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1	Cartridge, 5.56MM, Reference	1 OF 4	10524633
			NEXT HIGHER ASSEMBLY: Not applicable

CATEGORY	EXAMINATION OR TEST 1/	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE 2/ /INSPECTION METHOD
Critical					
1.	Case split in K, L or M location (6)		100%	3.1	Visual
2.	Case split in I, S or J location with loss of powder (6)		100%	3.1	Visual
3.	Perforated Case (7)		100%	3.1	Visual
4.	Primer missing (32)		100%	3.1	Visual
5.	Primer cocked (33)		100%	3.1	Visual
6.	Primer inverted (34)		100%	3.1	Visual
7.	Weight, under min 3/		100%	3.1	Balance
Major	(Class AQL - .25% except for Major 101)				
101.	Case split in I, S, or J location with no loss of powder (6)		100%	3.1	Visual
102.	Corroded or stained, (if etched case) (2)			3.1	Visual
103.	Round head, case (4)			3.1	Visual
104.	Dented case (5)			3.1	Visual

NOTES: 1/ Numbers after defect descriptions refer to visual defect standards in MIL-STD-636 (NATO Caliber 7.62mm Section).
2/ Refer to MIL-STD-636 (NATO Caliber 7.62mm Section) for visible defects standards. In the event of conflict between paragraph 4.3.2.1 of this specification and MIL-STD-636 as to defect classification the classification specified in paragraph 4.3.2.1 shall apply.
3/ Each lightweight cartridge shall be disassembled and the propellant weighted. Any cartridge containing less than 13 grains of propellant shall be classed as a critical defect. Any cartridge containing 13 grains or more of propellant shall be classed as a major defect.
4/ Detects other than those listed in MIL-STD-636 (7.62mm section).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46397C(AR)

PARAGRAPH	TITLE	SHEET 2 OF 4	DRAWING NUMBER
4.3.2.1	Cartridge, 5.56mm, Reference		10524633
			NEXT HIGHER ASSEMBLY: Not applicable

CATEGORY	EXAMINATION OR TEST 1/	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE 2/ /INSPECTION METHOD
105.	Draw scratch in case (8)			3.1	Visual
106.	Beveled underside of head (10)			3.1	Visual
107.	Case mouth not crimped in cannelure (11)			3.1	Visual
108.	Scaly metal on case (12)			3.1	Visual
109.	No chamfer on head (rim) (13)			3.1	Visual
110.	No visible evidence of mouth anneal (21)			3.1	Visual
111.	Split bullet jacket (24)			3.1	Visual
112.	Loose bullet (25)			3.1	Visual/Manual
113.	Missing cannelure			3.1	Visual
114.	Scaly metal on bullet (27)			3.1	Visual
115.	Loose primer (35)			3.1	Visual/Manual
116.	Total length			3.1	Gage
117.	Cartridge profile failure (requiring more than 10 lbs dead weight to insert in profile and alignment gage)			3.1	Gage
118.	Diameter of extractor groove, max			3.1	Gage
119.	Diameter of head			3.1	Gage
120.	Thickness of head			3.1	Gage
121.	Length to shoulder datum			3.1	Gage
122.	Depth of primer			3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1	Cartridge, 5.56mm, Reference	3 OF 4	10524633
			NEXT HIGHER ASSEMBLY: Not applicable

CATEGORY	EXAMINATION OR TEST 1/	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE 2/ /INSPECTION METHOD
Minor	(Class AQL - 1.5%)				
201.	Discolored, dirty, oily, or smeared (waterproofing) (1)			3.1	Visual
202.	Dented case (5)			3.1	Visual
203.	Draw scratch in case (8)			3.1	Visual
204.	Scratch in case (9)			3.1	Visual
205.	Scaly metal on case (12)			3.1	Visual
206.	Fold, wrinkle, buckle or bulge in case (14, 15, 16, 17)			3.1	Visual
207.	Head stamp missing or illegible (18)			3.1	Visual
208.	Defective head (19)			3.1	Visual
209.	Defective mouth (20)			3.1	Visual
210.	Dented bullet (22)			3.1	Visual
211.	Bullet scratched (23)			3.1	Visual
212.	Scaly metal on bullet (27)			3.1	Visual
213.	Upset (crooked) point (28)			3.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46397C(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1	Cartridge, 5.56mm, Reference	4 OF 4	10524633
			NEXT HIGHER ASSEMBLY: Not applicable

CATEGORY	EXAMINATION OR TEST 1/	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE 2/ /INSPECTION METHOD
214.	Blunt point (30)			3.1	Visual
215.	Defective cannelure (31)			3.1	Visual
216.	Nicked or dented primer (36)			3.1	Visual
217.	No waterproofing material (primer pocket joint) (37)			3.1	Visual
				3.1	Visual
218.	Defective crimp (38)			3.1	Visual
219.	Diameter of extractor groove, min			3.1	Gage
220.	Poor workmanship 4/			3.9	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.3.3 Testing. Testing is described in the Quality Conformance Inspection tables (4.3.3.1). Testing shall be conducted in accordance with the methods and procedures specified in 4.3.

a. Test samples. The quantities for the various tests shall be as specified in paragraph 4.3.3.1. Only cartridges having met the visual and dimensional requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. Sufficient cartridges shall be selected such that all testing (including testing of second samples where applicable) can be performed. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

b. Firing defects and associated acceptance-rejection criteria are specified in Table I.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46397C(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.3.1	Cartridge, 5.56mm, Reference	1 OF 2	10524633
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	TESTS				
	Bullet extraction	25	1/	3.2	4.3.1
	Residual stress	50	1/	3.3	4.3.2
	Waterproofness	50	2/	3.4	4.3.3
	Velocity	300	3/	3.5	4.3.4
	Chamber pressure 4/	300	3/	3.6	4.3.5
	Port pressure 4/	300	3/	3.7	4.3.6
	Function and casualty	600	5/	3.8	4.3.7
	Hardness			3.1	4.4.8
	Head	10	3/	3.1	4.4.8.1
	Sidewall	10	1/	3.1	4.3.8.2

NOTES:

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 10524633
4.3.3.1	Cartridge, 5.56mm, Reference				NEXT HIGHER ASSEMBLY Not applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.				
	2/ Failure of ten or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than three but less than ten cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sampe may be tested. The lot shall be rejected if, in the combined first and second sample, ten or more cartridges fail to comply with the applicable requirement.				
	3/ Failure of the cartridges to comply with the applicable requirements shall be cause for rejection of the lot.				
	4/ Chamber pressure and port pressure shall be fired simultaneously.				
	5/ Examination for firing defects shall be made during the velocity and pressure tests. The lot shall be rejected when firing defects exceed the acceptance number in Table III.				

NOTES: 6/ See Table I for defect classification and accept/reject criteria.

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TABLE I

Firing defect classification and accept-reject criteria

Defect Class	Defect Description	ACC	REJ
	CARTRIDGE FAILURES		
Critical	a. Bullet in bore	0	1
Critical	b. Misfire - Vent hole missing or blocked 1/	0	1
Major	c. Misfire - (excluding b. above)	1	2
	Primer Failures		
Major	a. Perforation of firing pin ident in primer cup	1	2
Major	b. Escape of gas through primer cup (excluding a above	1	2
Major	c. Loose primer 2/	2	3
Major	d. Blown primer 2/	0	1
Major	e. Dropped primer 2/	0	1
Major	f. Escape of gas around primer cup 3/		
Major	(1) 50% or more than 50% of periphery	6	7
Minor	(2) Less than 50% of periphery	12	13
	Case Casualties 4/		
1. Longitudinal split			
Major	a. Body (J)	2	3
Major	b. Body (K)	1	2
Major	c. to head (L)	0	1
Major	d. Through head (M)	0	1
Minor	e. Neck and shoulder (I or S)	11	12

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II. Circumferential Rupture

Major	a. Partial, shoulder or body (J & S)	1	2
Major	b. Partial, body (K)	0	1
Major	c. Partial, head (L)	0	1
Major	d. Complete	0	1

Note 1/ Each cartridge that misfires shall be disassembled and examined for a blocked or absent vent hole. If the vent hole is blocked or missing the lot shall be rejected.

Note 2/ Primer Defect Description:

a. Loose primer. The fired primer shall be called loose when it protrudes above the case head or movement of the primer can be observed by applying pressure on the primer by means of a probe, thumbnail or fingertip.

b. Blown primer. A primer which, when the cartridge is fired, is separated completely from the head of the cartridge case, and both the head of the case and the primer pocket are grossly distorted and deformed. The severity of this condition is such that it is readily detectable with the naked eye.

c. Dropped primer. A primer which falls from the primer pocket after the cartridge is fired.

Note 3/ For location of defects indicated by letters in parenthesis, see Drawing 7643674.

4.2.4 Packaging, packing, and marking. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing or as required by the contract. During or immediately prior to the packaging operation, 100 percent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test cartridge shall be classed as a critical defect. Occurrence of any incorrect type other than high pressure test shall be classed as a major defect. All non-conforming cartridges shall be rejected.

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4.2.5 Inspection equipment. The inspection equipment required to perform the examination and test prescribed herein is described in the Paragraph Reference/Inspection method column in the tables starting with Paragraph 4.3.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-480 78 and 6.2 herein.

4.2.5.1 Inspection equipment list. The examination and tests shall be made using equipment listed on LI-10524633, except as specified in 4.3.

4.3 Methods of inspection.

4.3.1 Bullet extraction. The test shall be conducted in accordance with SCATP-5.56. The rate of travel of the test head shall be not less than three nor more than six inches per minute.

4.3.2 Residual stress (Mercurous nitrate). The test shall be conducted in accordance with SCATP-5.56.

4.3.3 Waterproof. The test shall be conducted in accordance with SCATP-5.56.

4.3.4 Velocity. Prior to this test, cartridges shall be contitioned at 70°F ± 2°F for two hours minimum. The test shall be conducted in accordance with SCATP-5.56 and the following: The five velocity barrel assemblies specified in paragraph 2.5.1.2 of the SCATP shall be selected from a quantity of eight barrel assemblies. Selection of the five barrel assemblies shall be made by firing 30 reference cartridges in each barrel. The five test barrel assemblies that have the lowest standard deviation for velocity shall be selected for the test. Prior to firing of the 30 reference cartridges, each barrel assembly shall be "broken in" by firing 100 standard cartridges. Twenty cartridges shall be fired in each of five velocity barrel assemblies. The average velocity and standard deviation of velocities for each velocity barrel assembly and the average velocity for the five assemblies shall be computed. If the average velocity for any velocity barrel assembly varies from the average for the five assemblies by more than plus or minus 14 feet/second on the first day of firing, then that assembly shall be discarded and another selected and firing conducted therein until the aforementioned condition is met by all five assemblies; thereafter, on the two succeeding days, the same five assemblies shall be used. An additional 20 cartridges shall be fired in the same manner in the same velocity barrel assemblies on each of two succeeding days. The average velocity and standard deviation of velocities for each 20-round test, the daily average velocity (20-rounds in each of five assemblies), and the grand average for the entire 15 velocity series (five assemblies on each of three days) shall be computed.

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4.3.5 Chamber pressure. Prior to this test, cartridges shall be contitioned at 70°F ± 2°F for two hours minimum. The test shall be conducted simultaneously with the port pressure test in accordance with SCATP-5.56 and the following: The five chamber (and port pressure) barrel assemblies specified in paragraph 2.5.1.2 of the SCATP shall be selected from a quantity of eight barrel assemblies. Selection of the five barrel assemblies shall be made by firing 30 reference cartridges in each barrel. The five test barrel assemblies that have the lowest standard deviation for chamber pressure shall be selected for the test. Prior to firing of the 30 reference cartridges, each barrel assembly shall be "broken in" by firing 100 standard cartridges. Twenty cartridges shall be fired in each of five pressure barrel assemblies on each of three days. The average pressure and standard deviation of pressures for each 20-round test, the grand average pressure for the entire 15 pressure series (five assemblies on each of three days) and the average of the standard deviations for the 15 pressure series shall be computed.

4.3.6 Port pressure. The test shall be conducted in accordance with SCATP-5.56 simultaneously with the chamber pressure test. Upon completion, the average port pressure for the entire series (five assemblies on each of three days) shall be computed.

4.3.7 Function and casualty. This test shall be conducted simultaneously with the velocity and pressure tests.

4.3.8 Hardness testing. The bullets shall be extracted, the propellant removed and the primers extracted. Each cartridge case of the sample shall be prepared and placed on the appropriate test fixture for testing in accordance with ASTM E92, Standard Test Method for Vickers Hardness of Metallic Materials.

4.3.8.1 Case head. The sectioned halves of each sample case shall be marked in such a manner that both halves can be positively identified as being from the same case. The individual hardness value for each prescribed point on each sample case head shall be recorded. Any value failing to meet the drawing requirement at a prescribed point(s) shall be cause for remeasurement of hardness of the point(s) at the mirror image position on the second half of the sectioned head. The higher of the two measurements shall be recorded as the value of record for determination of conformance to drawing requirements.

4.3.8.2 Case sidewall. The average of the hardness values of the sample cases for each prescribed point along the sidewall exterior surface shall be computed and charted in accordance with the drawing requirements.

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4.3.9 Defect penalty. In any ballistic test in which the occurrence of a firing defect prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the acceptance sample shall be penalized for such defects in accordance with Table I.

5. PACKAGING

5.1 Packaging - level A (worldwide shipment). The cartridges shall be packaged in accordance with Drawing 10523634 and in accordance with Code of Federal Regulations Title 49, Parts 100-199.

5.2 Packing - level A (worldwide shipment). Packing shall be in accordance with Drawing 10523635 and in accordance with Code of Federal Regulations, Title 49, Parts 100-199.

5.3 Marking and labeling. Marking and labeling of the packed cartridges shall be in accordance with Drawing 10523635 and in accordance with Code of Federal Regulations, Title 49, Parts 100-199.

6. NOTES

6.1 Intended use. The cartridges covered by this specification are intended for use as reference cartridges for calibration purposes in 5.56MM cartridge acceptance testing.

6.2 Ordering data. Ordering data should include:

a. Title, number and date of this specification.

b. Type and level of packing for the cartridges.

c. Provisions for the submission of Inspection Equipment Designs (DI-R-10054) (see 6.3).

d. Provisions for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government (see 6.6).

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078). Submit inspection equipment designs as required to Commander, ARDC, ATTN: AMSMC-QAF-I (D), Dover, NJ 07801-5001.

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6.4 Hazard notice. The cartridge described herein and some of its components are flammable and explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to prevent fire, explosion, adverse environment, rough handling, corrosive atmosphere, and electrically inducted incidents. Such measures shall include the employment of an effective safety program that addresses the inherent hazards associated with the cartridge.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawing prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, US Army Armament Research and Development Command (ARRADCOM) or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.6 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by the US AMCCOM, on one (1) copy of all ballistic data and the ammunition data card for each lot shall be forwarded to: Commander, ARDC, ATTN: AMSMC-QAF-S(D), Dover, NJ 07801-5001.

6.7 In-process controls.

6.7.1 Uniformity of charge. A machine shall be chosen which is known to drop charges with better than average uniformity and which gives constant results in the daily production tests. If relative machine performance is unknown, special tests should be fired to assure selection of a good machine.

6.7.2 Loading control. A quantity of 20 rounds shall be fired for pressure and velocity for each four hours of an eight hour shift to establish velocity and pressure values, and check charge weight and unformity of charge weight every hour. Once established, the charge weight shall remain constant throughout loading.

6.8 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issues due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-AB45)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-46397C	CARTRIDGE, 5.56MM, REFERENCE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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